

Key Findings

Overview

A total of **148,670** loan applications were analyzed. The portfolio recorded an overall **default rate of 24.64%**, providing the baseline for evaluating borrower risk across different segments.

Risk Tier Analysis

The engineered **Risk Score** effectively differentiated borrower risk. Loans classified as **High Risk** recorded an **18.6%** default rate compared with **14.0%** for **Low-Risk** borrowers, indicating a clear increase in risk across tiers.

Regional Analysis

The **North-East** region recorded the highest default rate at **30.45%**, while the **North** region had the lowest rate at **22.51%**. This suggests noticeable regional variation in loan performance.

Income Tier Analysis

Borrowers in the **High-Income** category recorded the highest default rate (**38.78%**), whereas the **Low-Income** group recorded **24.61%**.

Loan Purpose

Loan purpose **P2** recorded the highest default rate (**33.08%**), while **P4** recorded the lowest (**22.97%**). This indicates that loan purpose may influence lending outcomes and deserves further investigation.

Credit Type Analysis

Applications assessed through **EQUI** showed a significantly higher default (**99.99%**) than **CIB**, **EXP**, and **CRIF**, which were all close to **16%**. This unusual result suggests that the EQUI category should be investigated further for possible data imbalance or business-specific factors.

Data Preparation

The dataset was cleaned by handling missing values in **Interest Rate**, **Interest Rate Spread**, **Upfront Charges**, and **Property Value**. Additional analytical features, including **Risk Score**, **Risk Tier**, **LTV Risk Flag**, and **DTI Risk Flag**, were created to support portfolio risk analysis.